

Chapter 5: Matrix Inverses Worksheet

1. Are each of the following statements true or false?

- (a) $A = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$ is an inverse of $B = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$.
- (b) The $n \times n$ identity matrix is invertible for any n .
- (c) If A, B are invertible matrices, and $C = B^{-1}A$, then $C^{-1} = BA^{-1}$.
- (d) Any invertible matrix must be a square matrix.
- (e) If A, B, C are 3×3 matrices with $AB = AC$ and A is invertible, then $B = C$.

2. Given square matrices $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$.

- (a) Verify that $A^{-1} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{1}{4} \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$.
- (b) Compute AB, BA .
- (c) Compute $A^{-1}B^{-1}, B^{-1}A^{-1}$, verifying that $A^{-1}B^{-1}$ is the inverse of BA and $B^{-1}A^{-1}$ is the inverse of AB .
- (d) Compute B^T and verify that $(B^{-1})^T$ is the inverse of B^T .

3. Rewrite each system as a matrix equation, then solve the systems using the fact that

$$\begin{bmatrix} 1 & -1 & -4 \\ -1 & 2 & 4 \\ 1 & 3 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} -18 & -15 & 4 \\ 1 & 1 & 0 \\ -5 & -4 & 1 \end{bmatrix}.$$

$$(a) \quad \begin{cases} x - y - 4z & = 1 \\ -x + 2y + 4z & = 1 \\ x + 3y - 3z & = 2 \end{cases}$$

$$(b) \quad \begin{cases} x - y - 4z & = 3 \\ -x + 2y + 4z & = 6 \\ x + 3y - 3z & = -5 \end{cases}$$

4. Recall that O denotes the zero matrix, and I denotes the identity matrix.

(a) Find two 2×2 matrices A such that $A^2 = O$.

(b) Find three 2×2 matrices A such that $A^2 = I$.

(c) Find three 2×2 matrices A such that $A^2 = A$.

(d) Find a pair of 2×2 matrices A, B such that $AB = O$ but $BA \neq O$.